

Technical Data sheet

NM0824-I mCPlc based LoRa Concentrator card

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Preface

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Revision history

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1. Introduction

The Concentrator module is targeted for a huge variety of applications like Smart Metering, IoT and M2M applications. It is a multi-channel high performance transmitter/receiver module designed to receive several LoRa packets simultaneously using different spreading factors on multiple channels. The concentrator module can be integrated into a gateway as a complete RF front end Module. It provides the possibility to enable robust communication between a LoRa gateway and a huge amount of LoRa end-nodes spread over a wide range of distance. The concentrator needs a host system for proper operation. This host system can be a PC or MCU that will be connected to the concentrator via SPI / USB Interface.



Figure 1-1: Image of the Concentrator Board

The Concentrator is able to receive up to 8 LoRa® packets simultaneously sent with different spreading factors and also on different channels. This unique capability allows to implement innovative network architectures advantageous over other short range systems:

- End-point nodes (e.g; sensor nodes) can change frequency with each transmission in a random pattern. This provides vast improvement of the system robustness in terms of interfere immunity and radio channel diversity.

- End-point nodes can dynamically perform link rate adaptation based (by adapting their spreading factors) on their link margin without adding complexity to the protocol. There is no need to maintain a table of which end point uses which data rate, because all data is demodulated in parallel.
- The capacity of the air interface can be increased due to orthogonal spreading factors.
- Due to the high range a star topology can be used. This results in simple implementation avoiding complex network layers, wireless routers and additional network protocol traffic.

1.1. Key Features

- Compact size 30 mm x 51 mm , mPCIe 52 pins compliant form factor
- LoRa® modulation technology
- Frequency band 915 MHz Support
- Output power level upto +27 dBm
- Sensitivity down to -139 dBm @SF12 with SX1250 TX/RX front-end
- SPI and USB interface
- SX1302 base band processor
- High-speed 250 / 500 kHz LoRa demodulator
- Multi-SF 125 kHz LoRa® reception with Fine Timestamp
- 1 (G)FSK demodulator
- 2 x SX1250 and 1 x SX1261 Tx/Rx front-ends
- Supply voltage +3.3V and/or +5V
- RF interface optimized to 50 Ω
- Range above 11Km (Line of Sight)
- Range of several km in urban environment
- Status LEDs

1.2. Applications

- Smart Metering
- Wireless Star Networks
- Home, Building, Industrial automation
- Remote Control
- Wireless Sensors
- M2M, IoT
- Wireless Alarm and Security Systems
- LoRaWAN, etc.

2. LoRa Modulation Technique

The Concentrator uses Semtech's LoRa® spread spectrum modulation technique. This modulation, in contrast to conventional modulation techniques, permits an increase in link budget and increased immunity to in-band interference.

LoRa also provides significant advantages in both blocking and selectivity, solving the traditional design compromise between range, interference immunity and energy consumption.

Semtech's LoRa® technology transceivers support several bandwidth options and spreading factors ranging from 7 to 12. The spread spectrum LoRa® modulation is performed by representing each bit of payload information by multiple chips of information. The rate at which the payload information is sent is referred to as the nominal symbol rate (R_s), the ratio between the nominal symbol rate and chip rate is the spreading factor and represents the number of modulation symbols sent per bit of information. Note that the spreading factor must be normally known in advance on both transmit and receive sides of the radio link as different spreading factors are orthogonal to each other. Note also the resulting signal to noise ratio (SNR) required at the receiver input. It is the capability to receive signals with negative SNR that increases the sensitivity, so link budget and range, of the LoRa receiver.

3. Module Overview

An overview about designation of the key components are given by the following picture

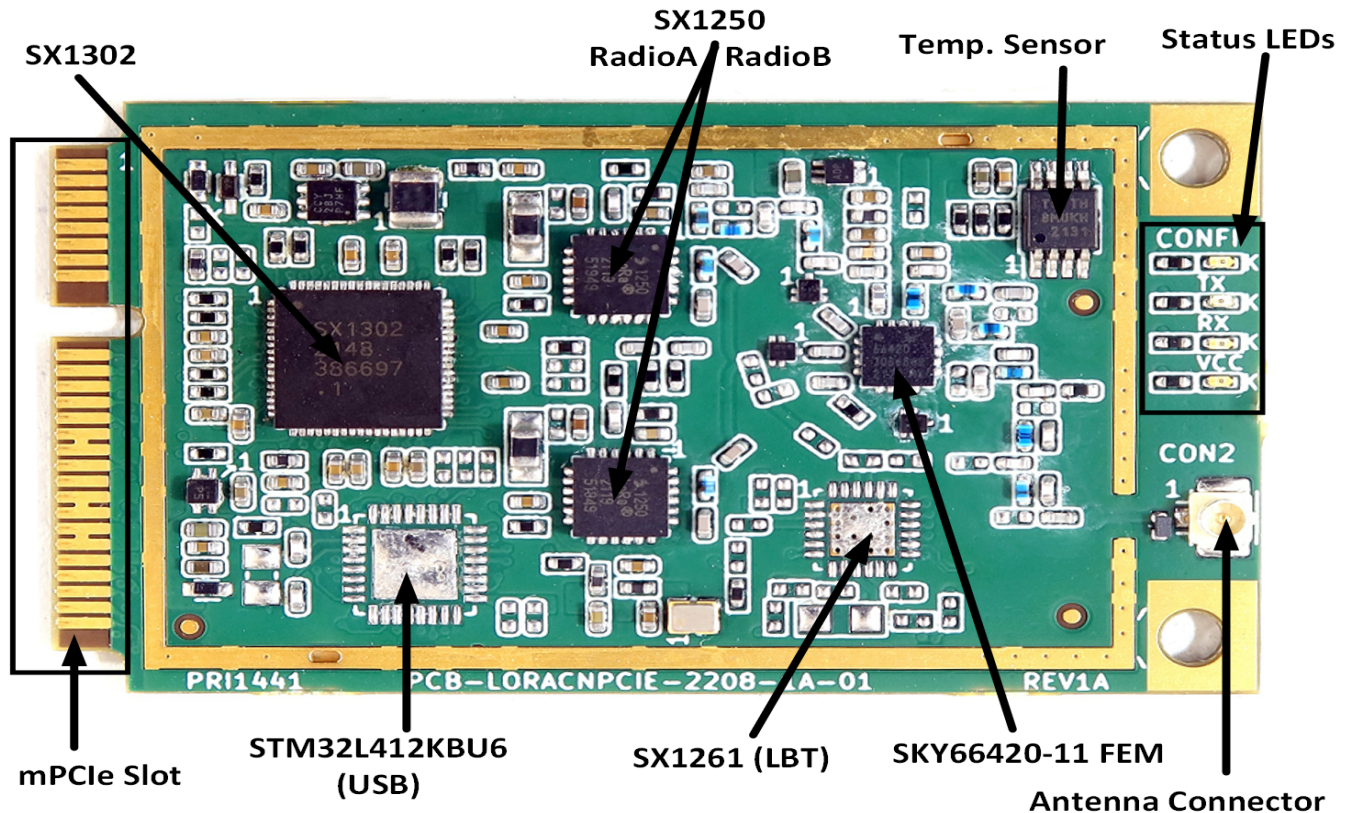


Figure 3-1: Component Overview of the Concentrator

NM0824-I Concentrator module is a new generation of LoRaWAN gateway module with mini-PCIe form-factor. Based on the Semtech® SX1302 baseband LoRaWAN® chip, NM0824-I unlocks the greater potential capacity of long-range wireless transmission for gateway products. It features higher sensitivity, less power consumption, and lower operating temperature. On board I2C Base Temperature Sensing available .

NM0824-I LoRaWAN gateway module has SPI and USB versions on both US915 frequency bands, It is designed for M2M and IoT applications and can be widely applied in LPWAN gateway supported scenarios. It would be a perfect choice for you to significantly reduce the technical difficulties and time-consumption when developing the LoRa gateway devices, including LoRaWAN gateway, etc.

4. Application Information

The design represents a compact reference implementation of the SX1302, along with its power management, clocks, Front End Module, RF matching and filtering.

4.1. Reference Block Diagram

The application block diagram is shown below:

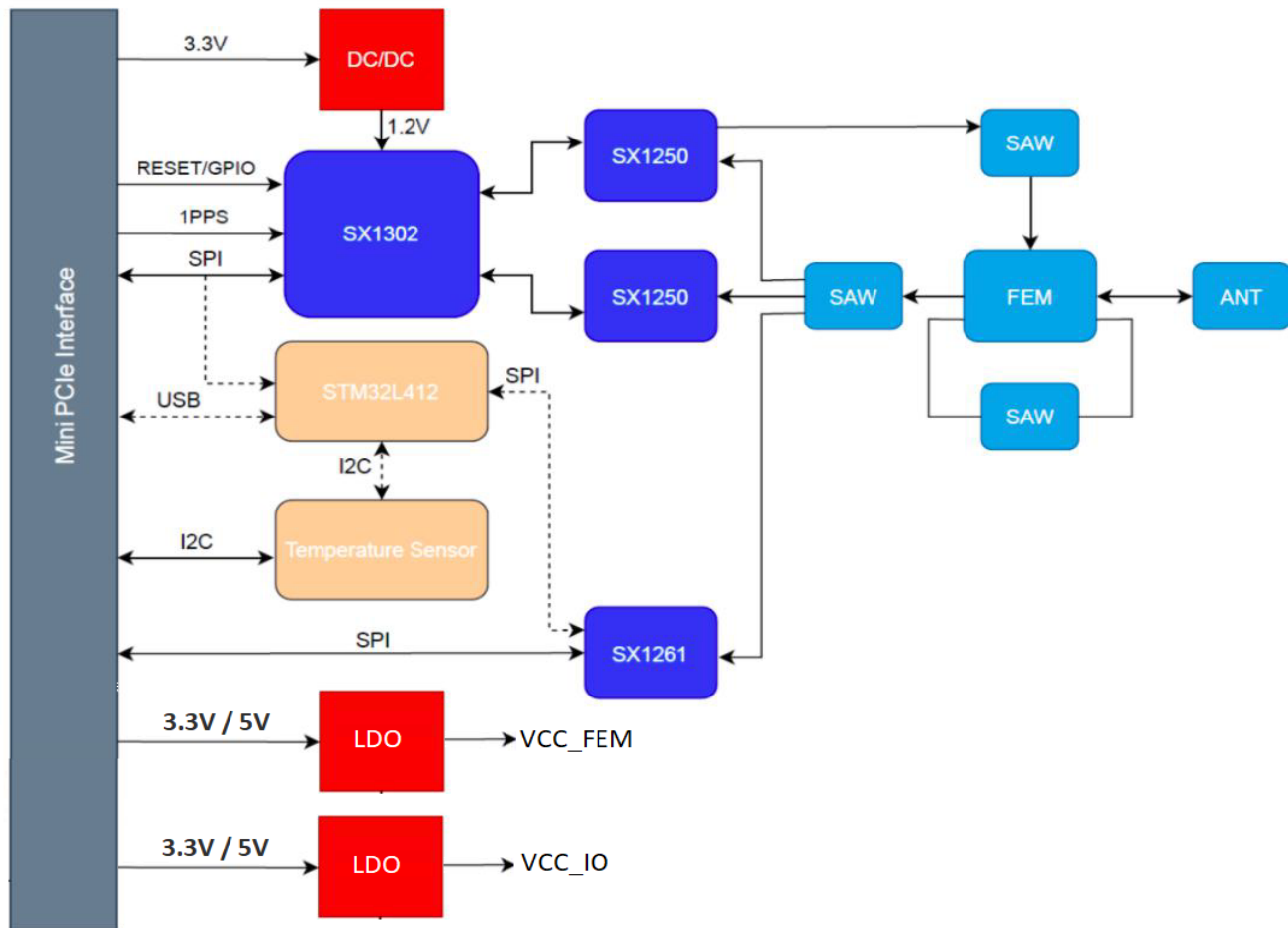


Figure 4-1-1: Application Design Block Diagram

5. Electrical Characteristics & Timing specifications

In the following different electrical characteristics of the Concentrator are listed. Furthermore details and other parameter ranges are available on request.

Note: Stress exceeding of one or more of the limiting values listed under “Absolute Maximum Ratings” may cause permanent damage to the radio module.

5.1. Absolute Maximum Ratings

Parameter	Condition	Min	Typ.	Max	Unit
Supply Voltage (VDD)		-0.3	3.3	3.6	V
Operating Temperature		-40		+85°C	°C

5.2. Global Electrical Characteristics

T = 25°C, VDD = 3.3V (typ.) if nothing else stated

Parameter	Condition	Min	Typ.	Max	Unit
Supply Voltage (VDD)		3.0	3.3	3.6	V

T = 25°C, VCC_IO = 3.3 V (typ.) if nothing else stated

Parameter	Condition	Min	Typ.	Max	Unit
Logic low input threshold (VIL)	"0" logic input	-0.3	-	0.3*VCC_IO	V
Logic high input threshold (VIH)	"1" logic input	0.7*VCC_IO	-	VCC_IO + 0.3	V
Logic low output level (VOL)	"0" logic output, I _{max} = 8mA	0	-	0.4	V
Logic high output level (VOH)	"1" logic output, I _{max} = -8 mA	VCC_IO – 0.6	-	VCC_IO	V

5.3. SPI Interface Characteristics

All timings are given in next table for Max load cap of 10 pF.

Parameter	Condition	Min	Typ.	Max	Unit
Timing constraints in SPI inputs					
tSPI,SCK	SCK period	100			ns
tSPI,SCKH	SCK high duration	40			ns
tSPI,SCKL	SCK low duration	40			ns
tSPI,SCKR	SCK rise time (10% VCC_IO ->90% VCC_IO)	0	2.5		ns
tSPI,SCKF	SCK fall time (90% VCC_IO ->10% VCC_IO)	0	2.5		ns
tSPI,DELAY	SCK lead time	40			ns
tSPI,QUIET	SCK trail time	40			ns
tSPI,CSH	Time between two successive CSN chip select	250			ns
tSPI,CSR	CSN rise time (10% VCC_IO ->90% VCC_IO)	0	2.5		ns
tSPI,CSF	CSN fall time (90% VCC_IO ->10% VCC_IO)	0	2.5		ns
tSPI,SETUP	Data in setup time	5			ns
tSPI,HOLD	Data in hold time	5			ns
SPI output timing specification					
tSPI,DOEN	SPI output enable time	0		10	ns
tSPI,DODIS	SPI output disable time	0		10	ns
tSPI,DOV	SCK out falling edge to MISO delay			15	ns

5.4. RF Characteristics

5.4.1. Transmitter RF Characteristics

The Concentrator has an excellent transmitter performance, which generally give a lot of possible settings for the power amplifier of the Concentrator. It is highly recommended, to use an optimized configuration for the power level configuration.

The Concentrator is specified for a max. RF output power of +27 dBm. Long-term operating of the Concentrator with more than +27 dBm can destroy the internal power amplifier of Concentrator. In Gateway output power configuration need to be ensured, that the maximum RF output power of +27 dBm is not exceeded.

Parameter	Condition	Min	Typ.	Max	Unit
Frequency Range	-	902	-	928	MHz
Modulation Techniques	FSK / LoRa™	-	+/-3	-	kHz
TX Frequency Variation vs. Temperature	-5°C to +55°C		+/-1		PPM
TX Power Variation vs. Temperature ¹	Max. power level, -5°C to +55°C		+/- 1		dB
	Max. power level, -40°C to +85°C		+/- 1		dB
TX Power Variation vs. Frequency	Max. power level		+/- 0.5		dB
TX Current Consumption	RX		40		mA
	Standby		7.5		mA
	TX (maximum power Level)		425		mA

Operational temperature range can be basically extended to -40°C to +85°C, but a larger power level drift vs. temperature need to be expected. In addition it is recommended, to use optimized TX settings which are also individually adapted to the temperature.

5.4.2. Receiver RF Characteristics

It is highly recommended, to use optimized RSSI calibration values. For both, Radio 1 and 2, the RSSI-Offset should be set to -169.

The following table gives typically sensitivity level of the Concentrator:

Signal Bandwidth/[kHz]	Spreading Factor	Sensitivity/[dBm]
125	12	-139
125	7	-125
250	12	-139
250	7	-125
500	12	-139
500	7	-125

6. Module Package

In the following the Concentrator module package is described. This description includes the Concentrator pinout as well as the modules dimensions.

6.1. Pinout

The Concentrator provides headers at the bottom side, which have a pitch of 2.54 mm. The description of the pins is given by Table below. An additional overview gives Figure 8-1-1.

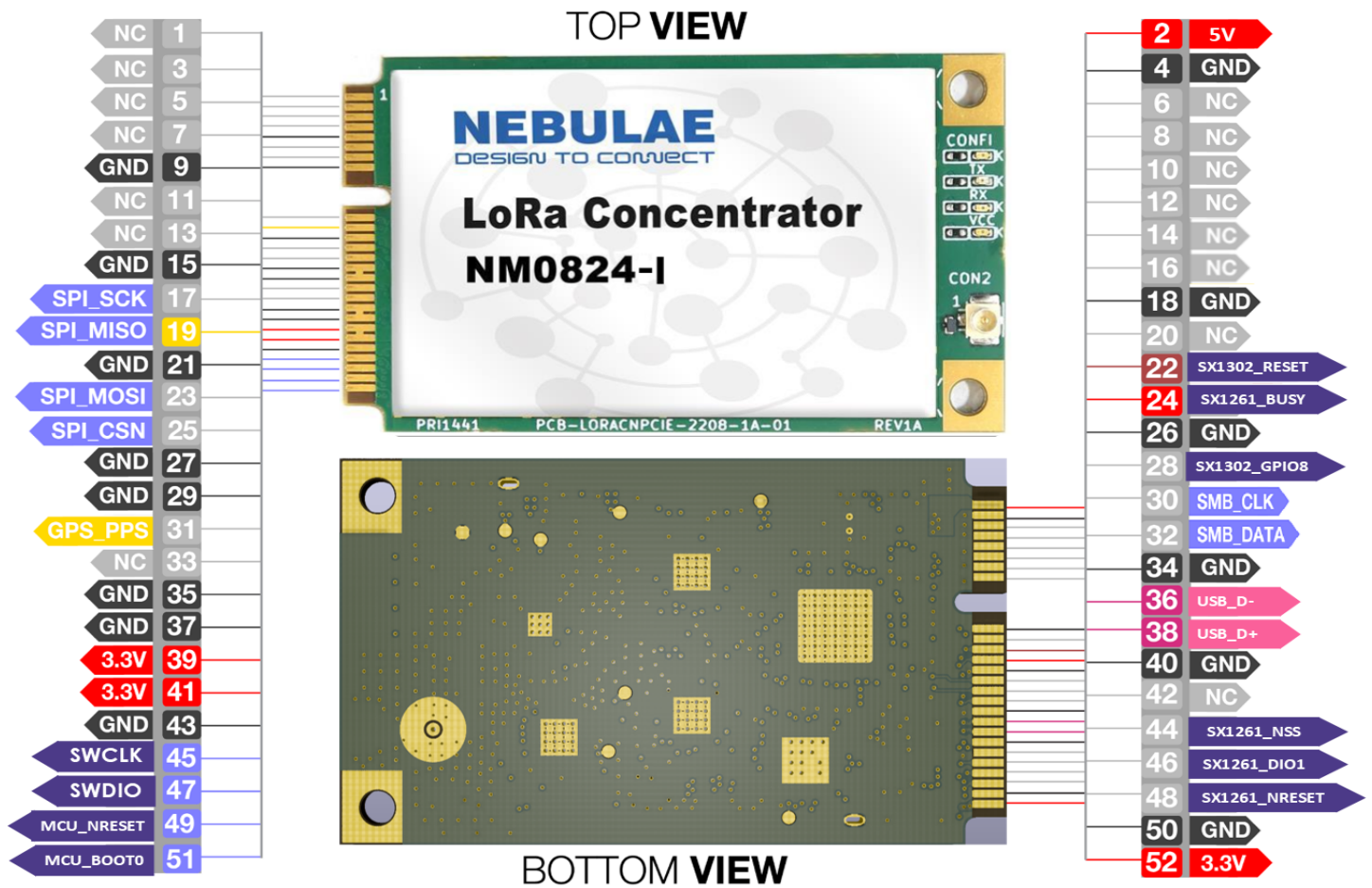


Figure 6-1-1 Concentrator outlines and pins of bottom connector in top view

Pin	Pin Name	Pin Type	Description
1	NC	NC	Reserved
2	VDD	Power	Optional +5.0V Supply Voltage(To be used if input supply to the board is +5.0V instead of +3.3V)
3	NC	NC	Reserved
4	GND	Power	System Ground
5	NC	NC	Reserved
6	NC	NC	Reserved
7	NC	NC	Reserved
8	NC	NC	Reserved
9	GND	Power	System Ground
10	NC	NC	Reserved
11	NC	NC	Reserved
12	NC	NC	Reserved
13	NC	NC	Reserved
14	NC	NC	Reserved
15	GND	Power	System Ground
16	NC	NC	Reserved
17	SPI_SCK	Input	SX1302 SPI-Clock
18	GND	Power	System Ground
19	SPI_MISO	Output	SX1302 SPI-MISO
20	NC	NC	Reserved
21	GND	Power	System Ground
22	SX1302_RESET	Reset	SX1302 Asynchronous Reset Input
23	SPI_MOSI	Input	SX1302 SPI-MOSI
24	SX1261_BUSY	O	SX1261 Busy indicator Output
25	SPI_CSN	Input	SX1302 SPI-Chip Select
26	GND	Power	System Ground
27	GND	Power	System Ground
28	SX1302_GPIO8	I/O	SX1303 General purpose IO
29	GND	Power	System Ground

30	SMB_CLK	Input	I2C Clock line for on-board temperature sensor
31	GPS_PPS	Input	Optional Time pulse (1PPS) from GPS
32	SMB_DATA	I/O	I2C Data line for on-board temperature sensor
33	NC	NC	Reserved
34	GND	Power	System Ground
35	GND	Power	System Ground
36	USB_D-	I/O	USB differential D+ (for USB to SPI Bridge)
37	GND	Power	System Ground
38	USB_D+	I/O	USB differential D+ (for USB to SPI Bridge)
39	VDD	Power	+3.3V Supply Voltage
40	GND	Power	System Ground
42	NC	NC	Reserved
43	GND	Power	System Ground
44	SX1261_NSS	Input	SX1261 SPI-Chip Select
45	SWCLK	I/O	STM32 Programing clock pin
46	SX1261_DIO1	I/O	SX1303 General purpose IO
47	SWDIO	I/O	STM32 Programing data IO pin
48	SX1261_NRESET	Reset	SX1261 Reset Input
49	MCU_NRESET	Reset	STM32 Reset Input
50	GND	Power	System Ground
51	MCU_BOOT0	I/O	MCU STM32 Boot mode pin.
52	VDD	Power	+3.3V Supply Voltage

Note: MISO-signal is always low impedance. Do not share with other MISO-signal by direct connection.

6.2 Module Dimensions

The outer dimensions of the Concentrator are given by 51.00 X 30.00 SQ mm $\pm$ 0.2 mm. The Concentrator provide two drills for screwing the PCB to another unit each with a drill diameter of 2.6 mm.

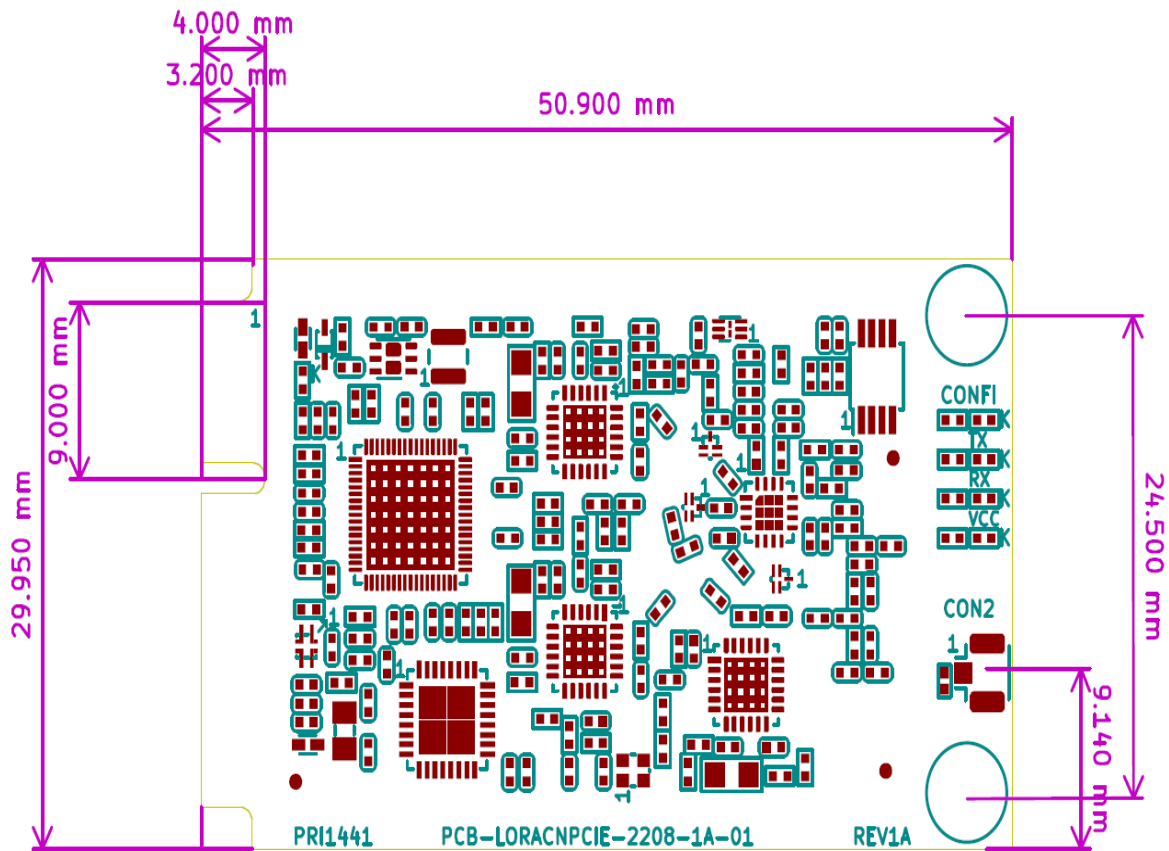


Figure 6-2-1: Concentrator outlines and pins of bottom connector in top view

7. Appendix

7.1. List of Abbreviations

GND	Ground
GPIO	General Purpose Input/Output
GPS	Global Positioning System
IF	Intermediate Frequency
IoT	Internet of Things
ISM	Industrial, Scientific and Medical
M2M	Machine to Machine
MCU	Microcontroller Unit
PCB	Printed Circuit Board
PPS	Pulse Per Second
RF	Radio Frequency
SNR	Signal to Noise Ratio
SPI	Serial Peripheral Interface

7.2. References

[1] Semtech, White Paper LoRa Modulation from www.semtech.com

[2] ERC Recommendation 70-03 “Relating to the use of Short Range Devices (SRD)”, Tromsø 1997, CEPT ECC subsequent amendments 13 October 2017

[3] Semtech, SX1302 Data Sheet from www.semtech.com